

Demand System Asset Pricing (DSAP): Structure, Methods, and Research Agenda

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Contents

1	Executive summary (what DSAP is and what I use it for)	3
2	Workshop map (3-session structure)	3
3	Data layer: what DSAP requires	3
3.1	Core objects and notation	3
3.2	Minimum data inputs	3
3.3	Holdings data construction (typical sources)	4
3.4	Investment universe and extensive margin	4
4	Model backbone: from micro foundations to equilibrium prices	4
4.1	Asset payoffs and a factor structure (generic template)	4
4.2	Investor problem (generic template)	4
4.3	A linear demand system representation	4
4.4	Market clearing and equilibrium prices	5
4.5	Reduced-form DSAP as a factor model (embeddings view)	5
5	Baseline empirical implementation: cross-section of U.S. equities	5
5.1	Characteristics-based demand (logit-like portfolio weights)	5
5.2	Identification: exogenous shifts in residual supply	5
5.3	Outputs I get from the estimated demand system	6
5.4	Counterfactuals via market clearing (computational template)	6
5.5	Decomposition and predictability (DSAP perspective)	6
6	Embeddings module: modern ML methods for DSAP	6
6.1	Why embeddings belong in DSAP	6
6.2	Canonical estimation targets	6
6.3	Representation learning approaches	7
6.4	Benchmarks (how I evaluate embeddings)	7
7	Fixed income application: upgrading credit pricing and risk assessment	7

8	Global demand system: exchange rates and asset prices	7
8.1	Objects	7
8.2	Nested logit demand (within and across asset classes)	7
8.3	Market clearing and equilibrium system	8
8.4	Decomposition and “specialness”	8
9	Research agenda (open questions)	8
10	Practical DSAP workflow (how I run a project)	8
10.1	Step-by-step template	8
11	Core reading list (what I cite and reuse)	9

1 Executive summary (what DSAP is and what I use it for)

Demand System Asset Pricing (DSAP) is a framework that uses *observed portfolio holdings* across investors, together with asset characteristics and supply, to estimate an empirical asset demand system. Market clearing then maps shifts in demand and supply into equilibrium prices and expected returns. In practice, I use DSAP to:

- quantify **demand elasticities** and **price impact** (liquidity through the lens of demand);
- recover **latent demand** (demand shifters) that are not explained by observables;
- run **counterfactuals** (policy, regulation, portfolio rebalancing, investor composition);
- build **asset and investor embeddings** using modern machine learning tools for valuation, risk, and classification;
- connect asset-pricing questions to **who holds what, who trades, and why prices move**.

2 Workshop map (3-session structure)

I organize the DSAP material into three modules, aligned with the workshop sessions:

1. Session 1: Demand systems for asset pricing

Motivation, micro foundations, data construction, model specification, identification, and estimation.

2. Session 2: Embeddings

Asset and investor embeddings (equities and fixed income), estimation approaches (recommender systems, representation learning), and applications.

3. Session 3: Applications and open questions

Counterfactuals, liquidity and price impact, variance decompositions, global demand systems, and a forward-looking research agenda.

3 Data layer: what DSAP requires

3.1 Core objects and notation

I use the following indexing convention:

- $i = 1, \dots, I$ investors (institutions, households, foreign sectors, central banks, etc.).
- $n = 1, \dots, N$ assets (stocks, bonds, or country–asset-class pairs in international applications).
- t time (quarterly in U.S. equity applications; annual in the global demand system example).

3.2 Minimum data inputs

At a high level, DSAP needs:

- **Holdings / portfolio weights:** $Q_{i,n,t}$ or $w_{i,n,t}$.
- **Prices:** $P_{n,t}$ (or market-to-book for equity; price per unit face value for debt).
- **Supply:** shares outstanding (equity) or quantities/face value (debt).
- **Asset characteristics:** $X_{n,t}$ (firm characteristics, bond features, macro/country characteristics).
- **Investor wealth / AUM:** $A_{i,t}$.

3.3 Holdings data construction (typical sources)

In practice, I treat the data problem as a *reconciliation* task:

- map holdings to a stable asset identifier (CUSIP/permno/ISIN);
- consolidate the investor entity (manager families, affiliates);
- align reporting frequency and reporting lags;
- enforce accounting identities where possible (market value = price $\times$ quantity).

Common examples across DSAP applications include:

- U.S. equity holdings for institutions (e.g., 13F-style filings) merged to CRSP/Compustat-style asset characteristics;
- mutual fund/ETF holdings and fixed income holdings from commercial providers;
- insurer bond holdings (e.g., regulatory schedules);
- international portfolio holdings and sectoral positions (including foreign exchange reserves in global applications).

3.4 Investment universe and extensive margin

A recurring DSAP detail is that investors do *not* consider all assets at all times. I represent this using an investment universe $N_{i,t}$ and impose

$$w_{i,n,t} = 0 \quad \text{if } n \notin N_{i,t}.$$

This structure is useful both empirically (sparse portfolios) and for identification (mandates generate quasi-exogenous variation in who can hold what).

4 Model backbone: from micro foundations to equilibrium prices

4.1 Asset payoffs and a factor structure (generic template)

I start from an asset-payoff representation with a factor structure:

$$D_{n,t+1} = \phi_{n,t} + \psi_{n,t}F_{t+1} + \nu_{n,t+1},$$

where F_{t+1} are common factors, and $\phi_{n,t}$ and $\psi_{n,t}$ can depend on observable characteristics $X_{n,t}$.

4.2 Investor problem (generic template)

A convenient benchmark is an investor problem with heterogeneous beliefs and risk aversion (and possibly constraints). I denote investor i 's choice as holdings $Q_{i,n,t}$ or portfolio weights $w_{i,n,t}$, subject to a budget constraint.

4.3 A linear demand system representation

A central DSAP object is a demand function that is *downward sloping in price* and depends on characteristics:

$$Q_{i,n,t} = -\pi_{i,t} P_{n,t} + \beta'_{i,t} X_{n,t} + \varepsilon_{i,n,t}. \quad (1)$$

Interpretation:

- $\pi_{i,t}$ is a (semi-)elasticity term that governs how sensitive demand is to price.
- $\beta_{i,t}$ maps observable characteristics into demand.
- $\varepsilon_{i,n,t}$ is **latent demand**: shifts in demand not explained by observables.

4.4 Market clearing and equilibrium prices

Let $S_{n,t}$ denote supply (shares outstanding or quantity). Market clearing implies

$$S_{n,t} = \sum_{i=1}^I Q_{i,n,t}.$$

Aggregating (1) over investors yields an equilibrium price representation:

$$P_{n,t} = \frac{1}{\pi_t} \left(\beta'_t X_{n,t} + \varepsilon_{n,t} - S_{n,t} \right), \quad \pi_t = \sum_{i=1}^I \pi_{i,t}, \quad (2)$$

so $\frac{1}{\pi_t}$ acts as an **equilibrium price-impact** term.

4.5 Reduced-form DSAP as a factor model (embeddings view)

A complementary, empirically convenient representation is a low-rank factor model for holdings:

$$Q_{i,n,t} \approx \lambda'_{i,t} x_{n,t} + u_{i,n,t}, \quad (3)$$

where $\lambda_{i,t} \in \mathbb{R}^K$ are **investor embeddings** and $x_{n,t} \in \mathbb{R}^K$ are **asset embeddings**. This turns DSAP into a high-dimensional representation learning problem while maintaining an equilibrium interpretation through market clearing.

5 Baseline empirical implementation: cross-section of U.S. equities

5.1 Characteristics-based demand (logit-like portfolio weights)

A widely used DSAP specification models portfolio weights via a multinomial-choice structure with an outside asset:

$$w_{i,t}(n) = \frac{\delta_{i,t}(n)}{1 + \sum_{m=1}^N \delta_{i,t}(m)}, \quad \delta_{i,t}(n) = \exp \left(b_{0,i,t} me_t(n) + b'_{1,i,t} x_t(n) + b_{K,i,t} \right) \epsilon_{i,t}(n),$$

where $me_t(n)$ is log market equity and $\epsilon_{i,t}(n) \geq 0$ is (multiplicative) latent demand.

The estimating equation is typically expressed in relative form:

$$\log \left(\frac{w_{i,t}(n)}{w_{i,t}(0)} \right) = b_{0,i,t} me_t(n) + b'_{1,i,t} x_t(n) + b_{K,i,t} + \log \epsilon_{i,t}(n).$$

5.2 Identification: exogenous shifts in residual supply

Identification requires a source of variation that shifts the residual supply faced by a given investor but is excluded from that investor's unobserved demand shifters. A core DSAP strategy uses **investment mandates** / **investment universes** (who is “allowed” to hold what) to construct instruments for endogenous price/size variables.

5.3 Outputs I get from the estimated demand system

Once I estimate the demand system, I can recover:

- **demand elasticities** (by investor and in aggregate);
- **latent demand** (extensive and intensive margins, depending on the specification);
- **equilibrium price impact** as a function of elasticities and the wealth distribution;
- **counterfactual prices** under alternative supply, investor composition, or demand shifts.

5.4 Counterfactuals via market clearing (computational template)

Market clearing often defines an implicit fixed point for the log price vector p :

$$p = f(p) \equiv \log \left(\sum_{i=1}^I A_i w_i(p) \right) - s,$$

so the equilibrium can be computed by iterating a fixed-point method (often with Jacobian approximations when N is large).

5.5 Decomposition and predictability (DSAP perspective)

A useful DSAP organizing identity is that equilibrium prices can be written as a function of state variables:

$$p_t = g(s_t, x_t, A_t, b_t, \epsilon_t).$$

This enables variance decompositions (supply- versus demand-side contributions) and a demand-based approach to predictability: if ϵ_t mean-reverts and other components are persistent, then high latent demand today can imply low expected returns going forward.

6 Embeddings module: modern ML methods for DSAP

6.1 Why embeddings belong in DSAP

Holdings data are extremely high-dimensional (many assets, many investors). Embeddings offer a disciplined way to:

- compress holdings into a low-dimensional state (asset and investor vectors);
- improve out-of-sample performance for valuation/risk tasks;
- support interpretability through clustering and similarity.

6.2 Canonical estimation targets

Let H be a holdings-based target (e.g., log weights or standardized positions). A canonical recommender-style objective is:

$$\min_{\{\lambda_i\}, \{x_n\}} \sum_{(i,n) \in \Omega} (H_{i,n} - \lambda_i' x_n)^2 + \eta_\lambda \sum_i \|\lambda_i\|^2 + \eta_x \sum_n \|x_n\|^2,$$

where Ω denotes observed holdings entries.

6.3 Representation learning approaches

Depending on the application, I can treat portfolios as sequences (“tokens”) and apply:

- **matrix factorization / PCA-type methods** (fast, transparent baselines);
- **Word2Vec-style objectives** (learn co-holding structure);
- **Transformer models (masked prediction)** for portfolio completion and richer nonlinear structure.

6.4 Benchmarks (how I evaluate embeddings)

I evaluate embeddings on economically relevant tasks, such as:

- **relative valuation**: link embeddings to future returns or valuation ratios;
- **return comovement**: explain covariance structure and factor exposures;
- **managed portfolios**: construct portfolios using embedding signals and evaluate performance;
- **risk assessment**: map embeddings into default risk, spread changes, or volatility (fixed income).

7 Fixed income application: upgrading credit pricing and risk assessment

In fixed income, I use embeddings to learn a **firm representation** from holdings and security-level information, then apply it to:

- explain credit spreads and credit spread volatility (including within rating categories);
- improve risk assessment relative to coarse rating buckets;
- build an “embedding-augmented” rating system that can be mapped into regulatory capital frameworks.

8 Global demand system: exchange rates and asset prices

8.1 Objects

I index assets by issuer country n and asset class $l \in \{S, L, E\}$: short-term debt (S), long-term debt (L), and equity (E). Prices and exchange rates jointly clear a large system of markets.

8.2 Nested logit demand (within and across asset classes)

A flexible DSAP specification uses a nested logit structure:

$$w_{i,t}(n, l) = w_{i,t}(n \mid l) w_{i,t}(l),$$

with

$$w_{i,t}(n \mid l) = \frac{\delta_{i,t}(n, l)}{1 + \sum_{m=1}^N \delta_{i,t}(m, l)}, \quad \log \delta_{i,t}(n, l) = \lambda_l \mu_{i,t}(n, l) + \Lambda_l x_{i,t}(n, l) + \epsilon_{i,t}(n, l),$$

and an “outer nest” across asset classes:

$$w_{i,t}(l) = \frac{\left(1 + \sum_{n=1}^N \delta_{i,t}(n, l)\right)^{\rho_l} \exp(\alpha_l + \xi_{i,t}(l))}{\sum_{k \in \{S, L, E\}} \left(1 + \sum_{n=1}^N \delta_{i,t}(n, k)\right)^{\rho_k} \exp(\alpha_k + \xi_{i,t}(k))}.$$

8.3 Market clearing and equilibrium system

Market clearing for each (n, l) typically takes the form

$$E_t(n) P_t(n, l) Q_t(n, l) = \sum_{i=1}^I A_{i,t} w_{i,t}(n, l),$$

so equilibrium jointly pins down exchange rates, short rates, long yields, and equity prices.

8.4 Decomposition and “specialness”

Once the global demand system is estimated, I can decompose variation in exchange rates and asset prices into:

- portfolio flows and quantities,
- macro fundamentals,
- latent demand (including reserve demand),

and quantify “specialness” (e.g., convenience yields) through counterfactual exercises that remove special latent demand and recompute equilibrium prices.

9 Research agenda (open questions)

Group open questions into six themes:

1. **Risk and comovement through holdings:** connect return comovement and factor structure to the joint distribution of holdings and investor shocks.
2. **Interpretable multimodal embeddings:** integrate holdings with text, characteristics, and market data to improve both performance and interpretation.
3. **Cross-asset and cross-country embeddings:** unify equity and bond embeddings and learn investor allocation structures across asset classes and geographies.
4. **Micro foundations for asset demand:** build models that deliver realistic sparsity, constraints, derivatives usage, and empirically plausible elasticities.
5. **Empirics and identification:** develop better benchmarks, richer substitution patterns, and more credible instruments for price endogeneity.
6. **Testing heterogeneous-agent models:** use DSAP estimates (elasticities and latent demand) as empirical targets for macro-finance models with heterogeneous intermediaries.

10 Practical DSAP workflow (how I run a project)

10.1 Step-by-step template

1. **Define the market and unit of observation:** equities, corporate bonds, sovereign debt, or global country-asset-class markets.
2. **Build the dataset:** holdings $\rightarrow$ weights/positions; merge prices, supply, and characteristics; validate accounting identities.

3. **Choose the demand specification:** characteristics-based logit/nested logit, linear demand, or low-rank reduced form.
4. **Identification strategy:** mandates/universes, quasi-experiments, policy-induced quantity shifts, gravity-style instruments, or granular IV.
5. **Estimate and validate:** in-sample fit, out-of-sample tasks, stability tests, and economic benchmark comparisons.
6. **Applications:** liquidity/price impact, decompositions, convenience yields, stress tests, valuation and risk systems.
7. **Counterfactuals and robustness:** solve market clearing under alternative scenarios; report sensitivity and interpretability checks.

11 Core reading list (what I cite and reuse)

References

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